

# JUNMING REN

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## EDUCATION

### Fudan University

Expected June 2027

*School of Data Science | Bachelor of Data Science and Big Data Technology*

- **Grade:** GPA: 3.62/4.0, TOEFL: 5.0
- **Honors:** Third Prize Scholarship of Fudan University, Honorable Prize in COMAP 2026

### University of Edinburgh

Sept. 2025 – Dec. 2025

*School of Informatics | Exchange Student*

## ACADEMIC RESEARCH

### Ethereum Arbitrage Transaction Detection

Feb. 2025 – Sep. 2026

*Submitted to the IEEE PacificVis 2027 TVCG Track; First Author and Lead Researcher*

*Conducted at the FDU-VIS Lab; Funded by FDUROP*

- Reconstructed dynamic control-flow graphs along actual transaction execution paths and applied structure-aware folding to reduce graph complexity while preserving key execution semantics.
- Recovered ERC-20 and ETH transfers directly from opcode-level EVM execution traces without relying on contract source code, ABIs, or event logs.
- Developed TraceWeaver, an interactive visual analytics system that links token-flow graphs, call trees, and control-flow graphs, enabling arbitrage fund flows to be traced to specific contract calls and opcodes.

### Minimizing Redundant Search Cost for Continuous Subgraph Matching

Feb. 2026 – May 2026

- Implemented query graph preprocessing to identify three types of substructures reused in the search process.
- Visualized the algorithmic workflow and experimental results and contributed to manuscript writing.

## WORK EXPERIENCE

### Shanghai Jiayicui Private Fund Management Co., Ltd.

May 2026 – Aug. 2026

*Quantitative Research Intern*

- Designed a three-stage deep learning model for convertible-bond return rank prediction, integrating a Transformer encoder, a GNN for neighborhood aggregation, and an MLP prediction head
- Incorporated equity nodes and constructed three types of correlation edges based on bond returns, bond volatility, and return correlations among the corresponding underlying stocks to enhance neighborhood aggregation.
- Developed a backtesting framework and iteratively improved the model using empirical results.

## SELECTED PROJECTS

- ERC-8004 Ecosystem Visualization | NFT Music: Wikipedia Article
- Vickey NFT Auction Contract | ThreeBody-ZHAOGE: Multi-Agent Dialogue Game

## TEACHING & CAMPUS SERVICE

### Teaching Assistant, Crypto Finance (English-Taught Graduate Course)

Sep. 2026 – Dec. 2026

- Designed a hands-on Sepolia-based midterm assignment and guided students through wallet setup, test-token acquisition, NFT minting, and marketplace sales; led in-class Q&A and assisted with final-paper reviews.

### Athletics Coordinator, School of Data Science

Sep. 2024 – Present

- Coordinated the School's registration and participation in university-wide athletic events.

## TECHNICAL SKILLS

**Programming:** Python (PyTorch, Polars), SQL, Solidity, C | **Tools:** Git,  $\text{\LaTeX}$